

Kareem Arab

Software Engineer, Applied AI & Voice AI | me@kareemarab.com | Ottawa, Canada | Open to relocating or remote

Applied AI engineer specializing in Voice AI, LLM evaluation, and production-grade AI systems. I build low-latency, reliable AI products across reasoning, telephony, backend infrastructure, and user experience.

Experience

Ada, Software Engineer, Voice AI, Nov 2024 - Present

- Led CPaaS integration for Ada's AI Voice Agent, redesigning telephony architecture to reduce call costs 85-95% while improving voice UX
- Built LLM evaluation pipelines and A/B testing workflows to measure voice agent quality, latency, and production behavior
- Delivered core modality-agnostic reasoning components powering conversational AI across voice and non-voice channels
- Designed long-silence handling and interaction patterns for more natural, resilient voice conversations

Amadeus, Software Engineer, Mar 2021 - Nov 2024

- Built authentication and MFA systems serving millions of users with 99.99% uptime requirements
- Led migration of authentication services from on-prem/GCP to Azure and Kubernetes, improving scalability and operational ownership
- Designed and ran A/B tests for product features, using experimentation data to guide product and engineering decisions
- Improved incident response practices and mentored junior engineers across delivery, reliability, and operations

Neurovine, Founding Software Engineer, Apr 2019 - Feb 2021

- Built HIPAA-compliant backend systems from zero to production using AWS Lambda and serverless infrastructure
- Integrated real-time ML inference into patient-facing concussion recovery workflows
- Built low-latency Bluetooth and data pipelines for wearable EEG devices with end-to-end encryption

Selected AI & Product Work

- **Brisa**: AI wealth platform with 1000+ users and \$55m assets tracked. #1 Fintech on Product Hunt. Go, React, AWS
- **SPC x OpenAI Hackathon**: Built an AI tool for reducing online polarization using NLP and GPT-based reasoning
- **Handshake**: Real-time iOS group chat and event platform using Swift, Firebase, and WebSockets
- **RP**: High-performance HTTP proxy with LRU caching. Python, multithreading

Technical Expertise

- **Applied AI**: Voice AI, LLM Agents, LLM Evals, RAG, LangChain, LangSmith, OpenAI, Anthropic
- **Voice Systems**: Telephony, CPaaS integrations, latency optimization, silence handling, conversational UX
- **Backend & Infra**: Python, Go, TypeScript, Kubernetes, Serverless, AWS, Azure, GCP, Redis, MongoDB
- **Product Engineering**: React, Next.js, iOS, A/B Testing, REST APIs, WebSockets

Research

IoT Lab @ Carleton University, Research Assistant, 2017 - 2019

- Conducted ML research for smart grid energy systems, resulting in 2 published conference papers
- Built real-time multi-modal data streaming systems using Raspberry Pi and IBM Cloud
- Mentored multiple 4th-year capstone projects

Education & Achievements

Carleton University, B.Sc. in Computer Science, Spring 2020

- Shopify Build Things Hackathon App Award (2018): Most innovative iOS app
- Published researcher: [Google Scholar Profile](#)